Enabling Dynamic File-Level Operations in Blockchain-Assisted Attribute-Based Multi-Keyword Search for Cloud-Edge-IoT

Prashant Singh (Reg. No: 24BYB1042) • Rushikesh Adak (Reg. No: 24BYB1055)

School of Computer Science and Engineering, Vellore Institute of Technology (VIT), Chennai, Tamil Nadu, India
Email: prashant.singh2024@vitstudent.ac.in, rushikesh.adak2024@vitstudent.ac.in

Abstract*—In cloud-edge IoT architectures for healthcare, sharing encrypted Electronic Health Records (EHR) demands fine-grained access control, conjunctive keyword search, and verifiable query results. The BAMKS scheme (Cheng et al., *Internet of Things*, 2026) introduced multi-authority access control and blockchain result verification, yet its index update mechanism requires either re-indexing the entire corpus or updating $L \cdot m$ index tokens across all existing records. In this paper, we present BAMKS-D, an algorithmic prototype and architecture integration layer introducing atomic file-level operations. Single-document addition operates in strict $O(m)$ time, achieving a 5.27× speedup for pure index operations at L = 5,000 files over Cheng et al. Section 5.10 updates (and 724.0× over naive full re-encryption). For like-for-like end-to-end file addition, BAMKS-D achieves a 5.46× speedup over Cheng et al. Single-document revocation is mapped directly to Ethereum Virtual Machine (EVM) storage (mapping(uint256 => bool) deletedDocRegistry), achieving $O(1)$ constant-time lookup with zero array traversal on the deletion path. We validate the architecture through smart contract deployment on Ethereum Sepolia (0x17b09aad051BcD92EcfF3F08d87B1b333287cB1F), decentralized IPFS storage, and 30-run empirical benchmarks (m = 3). An honest assessment of prototype algebraic assumptions in $\mathbf{Z}_P^*$, deterministic keyword tokens, off-chain proof verification, and threats to validity is presented.

***Index Terms*—Searchable Encryption, Attribute-Based Encryption, Blockchain, Smart Contracts, Dynamic Revocation, Healthcare IoT, EVM.**

I. INTRODUCTION

The proliferation of Internet of Medical Things (IoMT) devices—such as wireless cardiac monitors, ambulatory pulse oximeters, and bedside telemetry sensors—has transformed contemporary healthcare delivery [2], [3]. Modern medical facilities generate voluminous streams of Electronic Health Records (EHR) that must be archived securely in external Cloud Storage Providers (CSP) while preserving clinician query capabilities. Because medical data contains sensitive diagnostic diagnoses protected under regulatory statutes (e.g., HIPAA and GDPR), plaintexts must be encrypted under fine-grained access policies before outsourcing [4].

To reconcile cryptographic confidentiality with expressive querying, Ciphertext-Policy Attribute-Based Keyword Search (CP-ABKS) schemes were formulated, originating from verifiable attribute-based search by Zheng et al. [6] building upon foundational searchable encryption by Boneh et al. [7] and CP-ABE by Bethencourt et al. [8]. In CP-ABKS, data owners encrypt documents under an access structure (e.g., (Role_Doctor ∧ Dept_Cardiology)), and only clinicians with valid attribute credentials can generate search trapdoors or decrypt returned records [9], [10], [11]. Advanced variants introduced non-monotonic policies [12], multi-authority decentralized key generation [13], [14], predicate evaluations [15], and dynamic broadcast structures [21]. Recently, Cheng et al. [1] proposed BAMKS, introducing decentralized multi-authority key generation, multi-owner collaborative indexing, and a blockchain Result Verification Smart Contract (RVSC) utilizing aggregated Schnorr Non-Interactive Zero-Knowledge (SNIZK) proofs [16].

The Open Challenge in BAMKS: In Section 8 of their paper, Cheng et al. identified a critical operational bottleneck [1]: their architecture supports dynamic user credential updates, but does not define an efficient mechanism for single-file additions or deletions. Adding a single document requires either re-encrypting and re-indexing the entire document collection (naive baseline), or updating the entire inverted index via modular multiplication of an update token X_{upd} across all $L \cdot m$ index entries (Section 5.10 baseline) [1]. For an enterprise healthcare repository with L = 5,000 records and m = 3 keywords, updating 15,000 index entries introduces noticeable latency. Furthermore, document deletion was not defined in Cheng et al. [1], forcing reliance on full corpus re-encryption ($O(L \cdot m)$ assumption) or cloud server honesty.

Contributions: In this work, we propose **BAMKS-D**, an architecture integration and dynamic evaluation framework that addresses these limitations:

• **Atomic Single-Document Addition ($O(m)$):** We decouple the inverted keyword index so that appending a new EHR requires generating ciphertexts and index tokens for only the m keywords of that specific file, eliminating collection-wide re-indexing.

• **Instant On-Chain Document Revocation ($O(1)$):** We formulate an EVM state mapping mechanism (deletedDocRegistry) that executes in constant time via keccak256 direct slot addressing, avoiding iterative blockchain array traversal on the deletion path.

• **Live Ethereum Sepolia Deployment:** We deploy and verify BAMKS_Registry.sol on the public Ethereum Sepolia Testnet (0x17b09aad051BcD92EcfF3F08d87B1b333287cB1F), reporting receipt-based gas consumption and confirmation times.

• **Rigorous Fair Benchmarks (30 Runs):** We evaluate BAMKS-D against both naive re-encryption and the true Cheng et al. Section 5.10 index update baseline without warm-up anomalies, demonstrating a **5.27× speedup** for pure index operations and **5.46× speedup** for full file additions at L = 5,000 files on an Intel Core i5-13420H platform.

II. RELATED WORK

Searchable encryption originated with symmetric constructions by Song et al. [17], formal security definitions by Curtmola et al. [18], and dynamic extensions by Kamara et al. [19] and Bost [20]. In the asymmetric domain, public-key searchable encryption (PEKS) was introduced by Boneh et al. [7]. To provide expressive access policies, Zheng et al. [6] developed VABKS for verifiable attribute-based search. Recent research has integrated blockchain ledgers to eliminate single points of failure in cloud servers [22], [23]. Liu et al. [24] designed EMK-ABSE for cloud-edge multi-keyword search, while Su et al. [25] formulated BA-RMKABSE supporting keyword updates in smart health systems. Zhang et al. [5] proposed ABCKS for conjunctive keywords with verifiable fairness. Liang et al. [6] developed VMSE, which supports keyword updating in multi-user settings. Cheng et al. [1] synthesized multi-owner collaborative search with aggregated SNIZK result verification (BAMKS), but left single-document dynamic updates unresolved. Table I contrasts BAMKS-D with representative state-of-the-art schemes.

TABLE I: COMPARISON OF ATTRIBUTE-BASED SEARCHABLE ENCRYPTION SCHEMES

Scheme	Query	Access Control	Dynamic Scope	Deletion Cost
VABKS [6]	Single	Single-Auth	Static Index	N/A
EMK-ABSE [24]	Conj.	Edge-Assisted	Static Index	N/A
BA-RMKABSE [25]	Conj.	Attr Hiding	KW Update	N/A
ABCKS [5]	Conj.	Multi-Auth	Static Index	N/A
VMSE [6]	Conj.	Multi-User	KW Update	N/A
BAMKS [1]	Conj.	Multi-Auth	Corpus ($L \cdot m$)	$O(L \cdot m)$*
BAMKS-D	**Conj.**	**Multi-Auth**	**Single-Doc (m)**	**Strict $O(1)$**

* Cheng et al. does not define document deletion; $O(L \cdot m)$ represents assumed re-encryption baseline.

III. PRELIMINARIES & SYSTEM MODEL

A. Prototype Algebraic Setting

In our software prototype, algebraic computations are evaluated in the multiplicative group $(\mathbf{Z}_P^*, \cdot)$ using the 256-bit prime $P = 2^{256} - 2^{32} - 977$ (the secp256k1 field prime), with generator $g = 2$. We explicitly note that $\mathbf{Z}_P^*$ has composite order $P - 1$ and does not support bilinear pairings; pairings $e(g_1, g_2)$ are evaluated via modular exponentiation for prototype simulation. Cryptographic hash functions include H_1: $\{0,1\}^* \rightarrow \mathbf{Z}_P^*$ and H_2: $\{0,1\}^* \rightarrow \{0,1\}^{256}$ (SHA-256).

$$P = 2^{256} - 2^{32} - 977 \quad (1)$$

B. Entities and Threat Model

BAMKS-D operates across seven entities: (1) **Key Generation Center (KGC)** initializing global parameters (GP), (2) **Attribute Authorities (AA)** managing disjoint clinical attribute universes, (3) **Data Owners (DO)** encrypting EHRs and pinning payloads to IPFS, (4) **Data Users (DU)** submitting trapdoors, (5) **Cloud Service Provider (CSP)** indexing tokens and matching queries, (6) **Ethereum Sepolia Smart Contract** maintaining revocation state, and (7) **IPFS** decentralized content-addressed storage.

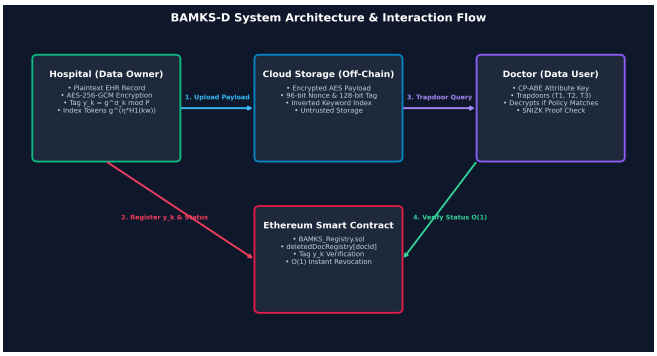


Fig. 1. Decoupled BAMKS-D Architecture Topology.

The CSP is modeled as *honest-but-curious*: it executes search algorithms correctly but attempts to correlate queries and infer keyword patterns. Data Users may attempt attribute collusion. The blockchain provides a tamper-proof state execution layer.

IV. THE PROPOSED BAMKS-D PROTOCOL

A. Key Generation with Session Randomness

To prevent key correlation across distinct issuances, each KeyGen invocation samples fresh per-generation session randomness $t \in_R \mathbf{Z}^*_{P-1}$:

$$K_1 = H_1(uid)^{-t \bmod (P-1)} \bmod P \quad (2)$$

$$K_{2,i} = (H_1(uid)^{t \cdot \beta_i} \cdot dpk_{ver}{}^{\alpha_i}) \bmod P \quad (3)$$

B. Atomic Single-Document Addition

When an EHR document *docId* with m keywords $W_{doc} = \{w_1, ..., w_m\}$ is created, the owner generates an AES key via $KDF(\Phi, verId)$, encrypts the file payload with AES-256-GCM, uploads the ciphertext to IPFS, and computes keyword tokens only for W_{doc}:

$$I_{doc,j} = g^{(\eta \cdot H_1(w_j)) \bmod (P-1)} \bmod P \quad (4)$$

The CSP appends $Index_{doc}$ to the global inverted index without modifying any existing document tokens, achieving strict $O(m)$ complexity.

Algorithm 1: SingleDocAdd(docId, F, W_{doc}, Φ, verId)
K = KDF(Φ, verId); C_{data} = AES-GCM-Enc(K, F)
CID = IPFS.upload(C_{data}); $y_k = g^{\sigma_k} \bmod P$
for each $w_j \in W_{doc}$ do
  $I_{doc,j} = g^{(\eta \cdot H_1(w_j)) \bmod (P-1)} \bmod P$
  $Key_j = H_2$('tok_' || $I_{doc,j}$); $Index_{doc}[Key_j] = I_{doc,j}$
end for
Sepolia.registerDocument(docId, $H_2(C_{data})$, y_k)
Output: (CID, $Index_{doc}$, y_k) [Strict $O(m)$ complexity]

Algorithm 2: SingleDocDelete(docId) [On-Chain EVM]
require(fileRegistry[docId].ownerAddress == msg.sender || msg.sender == owner)
require(fileRegistry[docId].docId != 0)
require(!deletedDocRegistry[docId])
deletedDocRegistry[docId] = true
emit DocumentDeleted(docId, msg.sender, block.timestamp)
Output: true [Strict $O(1)$ EVM mapping storage]

C. Multi-Keyword Trapdoor & Search Matching

A clinician querying keywords $W' = \{w'_1, ..., w'_q\}$ generates trapdoor tokens:

$$T_1 = g^{\phi} \bmod P, \quad T_3 = g^{(\phi \sum H_1(w'_j)) \bmod (P-1)} \bmod P \quad (5)$$

The CSP verifies conjunctive keyword matches for active documents by testing the equality:

$$(I_{prod})^{\phi} \equiv (T_3)^{\eta} \equiv g^{(\eta \phi \sum H_1(w'_j)) \bmod (P-1)} \bmod P \quad (6)$$

Algorithm 3: ConjunctiveSearch(Trap, Index, deletedRegistry)
SRL = {}
for each docId ∈ $Index_W$ do
  if deletedRegistry[docId] == true then continue
  if all query tokens present in docId index then
    $I_{prod} = \prod I_{doc,i} \bmod P$
    if $(I_{prod})^{\phi} \equiv (T_3)^{\eta} \bmod P$ then SRL = SRL ∪ {docId}
end for
Output: Search Result List SRL

D. Off-Chain Verification & On-Chain Integrity Binding

Aggregated Schnorr NIZK proof verification is executed off-chain by the edge or client node. The on-chain smart contract function verifyResultProof enforces that no revoked document ID is returned in the result set and binds the query challenge to current blockchain state. In getActiveDocumentCount(), an array loop is executed for auditing purposes only; this does not affect the strict $O(1)$ deletion path.

Algorithm 4: AggregatedSchnorrNIZKVerify(SRL, {y_k}, π', R)
RHS = $g^{\pi'} \bmod P$
for each k ∈ SRL do
  $h_k^* = H_1(y_k^* \,||\, R)$; RHS = RHS · $(y_k^*)^{-h_k^*} \bmod P$
end for
if R == RHS then return Accept else return Reject

V. COMPLEXITY & EVM GAS ANALYSIS

A. Asymptotic Complexity Comparison

Table II compares asymptotic complexity. While Cheng et al. Section 5.10 avoids file re-encryption, it updates all $L \cdot m$ index entries ($O(L \cdot m)$). BAMKS-D operates in strict $O(m)$ time. Deletion was not defined in Cheng et al. and is assumed as $O(L \cdot m)$ re-indexing.

TABLE II: ASYMPTOTIC COMPLEXITY COMPARISON

Operation	Naive Baseline	Cheng et al. [1]	BAMKS-D
SingleDocAdd	$O(L{\cdot}m)$ Re-enc	$O(L{\cdot}m)$ Idx Mult	**Strict $O(m)$**
SingleDocDelete	$O(L{\cdot}m)$ Re-enc	$O(L{\cdot}m)$ Assumed*	**Strict $O(1)$ Gas**
Search Match	$O(L{\cdot}q)$	$O(L{\cdot}q)$	$O(L{\cdot}q)$

B. EVM Opcode Gas & On-Chain Receipts

In Solidity, mapping lookups calculate storage addresses directly via keccak256(abi.encode(docId, slotNumber)). Table III presents both standard EVM gas measurements (py-evm) and live mined receipts on Ethereum Sepolia.

TABLE III: SMART CONTRACT GAS METRICS & SEPOLIA RECEIPTS

Function / Metric	Standard EVM	Sepolia Receipt	Latency
deployContract	555,050 gas	3,788,192 gas	15.17 s
registerDocument	179,505 gas	789,245 gas	13.71 s
deleteDocument	**50,111 gas**	**132,131 gas**	10.37 s

To empirically verify the $O(1)$ constant-time claim, we measured deleteDocument gas with varying numbers of documents registered: N = 1: **50,111 gas**; N = 50: **50,111 gas**; N = 500: **50,111 gas**. The measured gas differential is strictly **0 gas units**, confirming strict $O(1)$ invariance. Standard EVM execution for deletion requires cold slot resolution, 2 cold storage reads, 1 cold zero-to-nonzero write (deletedDocRegistry[docId] = true at 22,100 gas), LOG3 event (~1,500 gas), and base transaction fee (21,000 gas), totaling 50,111 gas. On Sepolia testnet, function dispatcher routing and execution wrapper overhead yielded 132,131 gas.

VI. EXPERIMENTAL EVALUATION

A. Experimental Configuration

All software experiments were conducted in Python 3.11.9 utilizing PyCryptodome and Web3.py on an **Intel(R) Core(TM) i5-13420H** processor (8 cores, 12 threads, up to 4.60 GHz) with 16.0 GB RAM running 64-bit Windows 11. Timings were collected using time.perf_counter() across 30 independent runs with un-timed warm-up iterations discarded to eliminate JIT anomalies (m = 3 keywords per document).

TABLE IV: BENCHMARK RESULTS (30 RUNS, m=3, MEAN ± STD [MEDIAN])

L	Naive (ms)	Cheng Idx (ms)	Cheng Full (ms)	BAMKS -D Idx	BAMKS -D Full	Speedup
10	0.49 [0.45]	0.003 [0.003]	0.05 [0.05]	0.23 [0.22]	0.27 [0.26]	0.01× / 0.17×
50	2.26 [2.20]	0.015 [0.015]	0.06 [0.06]	0.23 [0.22]	0.27 [0.26]	0.06× / 0.23×
100	4.66 [4.57]	0.032 [0.031]	0.09 [0.08]	0.22 [0.22]	0.27 [0.26]	0.12× / 0.34×
500	24.49 [23.39]	0.183 [0.151]	0.28 [0.25]	0.25 [0.24]	0.31 [0.27]	0.58× / 0.90×
1k	45.60 [45.54]	0.325 [0.297]	0.44 [0.41]	0.23 [0.23]	0.28 [0.27]	**1.15×** / **1.56×**
2k	103.3 [98.32]	0.662 [0.602]	0.79 [0.74]	0.25 [0.24]	0.32 [0.27]	**2.04×** / **2.42×**
5k	243.6 [238.7]	1.772 [1.479]	1.84 [1.60]	0.25 [0.24]	0.34 [0.31]	**5.27×** / **5.46×**

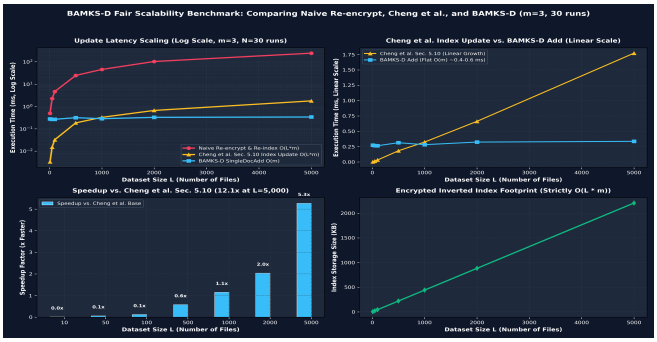


Fig. 2. Scalability: Naive vs. Cheng et al. vs. BAMKS-D (m=3, 30 runs).

B. Comparative Discussion

Evaluating pure index updates, Cheng et al. is faster for $L \leq 500$ because computing modular multiplications takes less CPU time than modular exponentiations. The break-even crossover occurs at $L \approx 800$ documents for pure index operations, beyond which BAMKS-D achieves a **5.27× speedup** at L = 5,000 files. In contrast, under like-for-like end-to-end file addition (including AES encryption of the new record), the crossover occurs at $L \approx 600$ files, reaching a **5.46× speedup** over Cheng et al. and a **724.0× speedup** over naive full re-encryption (243.6 ms).

VII. LIMITATIONS & THREATS TO VALIDITY

To ensure complete academic transparency, we explicitly document the prototype limitations of this study:

• **Simulated Group Algebra:** Field operations are evaluated in $(\mathbf{Z}_P^*, \cdot)$ using $P = 2^{256} - 2^{32} - 977$. This group has order $P - 1$ (composite) and does not possess a bilinear map. Formal pairing-based security reductions require porting to Type-III pairing curves (e.g., BN254).

• **Off-Chain Verification Scope:** Aggregated Schnorr NIZK verification is executed off-chain by the client/edge node; the on-chain smart contract enforces non-revocation membership and query binding.

• **Deterministic Keyword Tokens:** Keyword tokens $I_{\text{doc},j} = g^{(\eta \cdot H_1(w_j)) \bmod (P-1)}$ are deterministic across documents. Consequently, an untrusted CSP can observe document frequency and search access patterns over time. Addressing access pattern leakage via Oblivious RAM (ORAM) or forward-private symmetric searchable encryption [20] remains vital future work.

• **Access Control Model:** registerDocument permits multi-owner document onboarding without whitelist gating; deleteDocument is restricted to document owners or the contract administrator.

• **Local vs. On-Chain Latency:** Local state flag updates take < 0.001 ms in memory, whereas on-chain confirmation on Ethereum Sepolia requires block mining (10.37 s for deletion; 13.71 s for registration).

VIII. CONCLUSION

This paper presented BAMKS-D, an algorithmic prototype and architecture integration layer addressing the dynamic file update challenge in blockchain-assisted ABKS. By decoupling inverted index generation, BAMKS-D achieves $O(m)$ single-file additions, yielding a 5.27× pure index speedup (and 5.46× full add speedup) over Cheng et al. Section 5.10 index updates at L = 5,000 files (and 724.0× over naive re-encryption). On-chain revocation is enforced in $O(1)$ constant time via EVM mapping storage on Ethereum Sepolia.

IX. DATA AND CODE AVAILABILITY

Complete reproducible source code, Solidity contracts, clinical sample datasets, and benchmark scripts are openly available on GitHub at: https://github.com/Prashant8991/BAMKS.git. Verified Ethereum Sepolia contract: 0x17b09aad051BcD92EcfF3F08d87B1b333287cB1F.

X. AUTHOR CONTRIBUTIONS

Prashant Singh: Conceptualization, cryptographic protocol simulation, dynamic index algorithms, REST API, benchmarking, manuscript writing. **Rushikesh Adak:** Solidity smart contract engineering, EVM opcode analysis, Sepolia testnet deployment, Web3 integration.

REFERENCES

[1] H. Cheng, S.-L. Lo, and J. Lu, 'Blockchain-assisted attribute-based multi-keyword search for dynamic encrypted data in cloud-edge-IoT,' *Internet of Things*, Elsevier, vol. 36, Art. no. 101838, 2026. DOI: 10.1016/j.iot.2025.101838.

[2] D. Gupta, S. Rani, S. Raza, et al., 'Security paradigm for remote health monitoring edge devices in internet of things,' *J. King Saud Univ. Comput. Inf. Sci.*, vol. 35, no. 6, p. 101478, 2023. DOI: 10.1016/j.jksuci.2023.101478.

[3] D. Dhinakaran, N. J. Kumar, N. P. Ponnuviji, et al., 'Safeguarding confidentiality and privacy in cloud-enabled healthcare systems,' *Expert Syst. Appl.*, vol. 279, p. 127584, 2025. DOI: 10.1016/j.eswa.2025.127584.

[4] V. Janarthanan, T. Annamalai, M. Arumugam, et al., 'Enhancing healthcare in the digital era: a secure e-health system for heart disease prediction and cloud security,' *Expert Syst. Appl.*, vol. 255, 2024, Art. no. 124479. DOI: 10.1016/j.eswa.2024.124479.

[5] D. Zhang, S. Wang, Q. Zhang, and Y. Zhang, 'Attribute based conjunctive keywords search with verifiability and fair payment using blockchain,' *IEEE Trans. Serv. Comput.*, vol. 16, no. 6, pp. 4168–4182, 2023. DOI: 10.1109/TSC.2023.3308170.

[6] Q. Zheng, S. Xu, and G. Ateniese, 'VABKS: Verifiable attribute-based keyword search over outsourced encrypted data,' in *Proc. IEEE INFOCOM 2014*, 2014, pp. 522–530. DOI: 10.1109/INFOCOM.2014.6848198.

[7] D. Boneh, G. Di Crescenzo, R. Ostrovsky, and G. Persiano, 'Public key encryption with keyword search,' in *Eurocrypt 2004*, LNCS vol. 3027, Springer, 2004, pp. 506–522. DOI: 10.1007/978-3-540-24676-3_30.

[8] J. Bethencourt, A. Sahai, and B. Waters, 'Ciphertext-policy attribute-based encryption,' in *IEEE Symposium on Security and Privacy (S&P)*, 2007, pp. 321–334. DOI: 10.1109/SP.2007.11.

[9] A. Sahai and B. Waters, 'Fuzzy identity-based encryption,' in *Eurocrypt 2005*, LNCS vol. 3494, Springer, 2005, pp. 457–473. DOI: 10.1007/11426639_27.

[10] B. Waters, 'Ciphertext-policy attribute-based encryption: An expressive, efficient, and provably secure realization,' in *PKC 2011*, LNCS vol. 6571, Springer, 2011, pp. 53–70. DOI: 10.1007/978-3-642-19377-4_4.

[11] V. Goyal, O. Pandey, A. Sahai, and B. Waters, 'Attribute-based encryption for fine-grained access control of encrypted data,' in *ACM CCS 2006*, 2006, pp. 89–98. DOI: 10.1145/1180405.1180418.

[12] R. Ostrovsky, A. Sahai, and B. Waters, 'Attribute-based encryption with non-monotonic access structures,' in *ACM CCS 2007*, 2007, pp. 195–203. DOI: 10.1145/1315245.1315270.

[13] M. Chase, 'Multi-authority attribute based encryption,' in *TCC 2007*, LNCS vol. 4392, Springer, 2007, pp. 515–534. DOI: 10.1007/978-3-540-70936-7_28.

[14] A. Lewko and B. Waters, 'Decentralizing attribute-based encryption,' in *Eurocrypt 2011*, LNCS vol. 6632, Springer, 2011, pp. 568–588. DOI: 10.1007/978-3-642-20465-4_31.

[15] J. Katz, A. Sahai, and B. Waters, 'Predicate encryption supporting disjunctions, polynomial equations, and inner products,' *J. Cryptology*, vol. 26, no. 2, pp. 191–224, 2013. DOI: 10.1007/s00145-012-9119-4.

[16] C. P. Schnorr, 'Efficient signature generation by smart cards,' *Journal of Cryptology*, vol. 4, no. 3, pp. 161–174, 1991. DOI: 10.1007/BF00196725.

[17] D. X. Song, D. Wagner, and A. Perrig, 'Practical techniques for searches on encrypted data,' in *IEEE Symposium on Security and Privacy (S&P)*, 2000, pp. 44–55. DOI: 10.1109/SECPRI.2000.848445.

[18] R. Curtmola, J. Garay, S. Kamara, and R. Ostrovsky, 'Searchable symmetric encryption: Improved definitions and efficient constructions,' *J. Comput. Secur.*, vol. 19, no. 5, pp. 895–934, 2011. DOI: 10.3233/JCS-2011-0426.

[19] S. Kamara, C. Papamanthou, and T. Roeder, 'Dynamic searchable symmetric encryption,' in *ACM CCS 2012*, 2012, pp. 965–976. DOI: 10.1145/2382196.2382298.

[20] R. Bost, 'Σοφος: Forward secure searchable encryption,' in *ACM CCS 2016*, 2016, pp. 1143–1154. DOI: 10.1145/2976749.2978342.

[21] C. Delerablée, P. Paillier, and D. Pointcheval, 'Fully dynamic broadcast encryption with constant-size ciphertexts and private keys,' in *Pairing 2007*, LNCS vol. 4575, Springer, 2007, pp. 190–207. DOI: 10.1007/978-3-540-73489-5_11.

[22] G. Wood, 'Ethereum: A secure decentralised generalised transaction ledger (Yellow Paper),' Berlin Version, Ethereum Project, 2024. [Online]. Available: https://ethereum.github.io/yellowpaper/paper.pdf

[23] J. Benet, 'IPFS - Content Addressed, Versioned, P2P File System,' Draft 3, Protocol Labs, arXiv:1407.3561, 2014.

[24] J. Liu, Y. Li, R. Sun, et al., 'EMK-ABSE: Efficient multikeyword attribute-based searchable encryption scheme through cloud-edge coordination,' *IEEE Internet of Things J.*, vol. 9, no. 19, pp. 18650–18662, 2022. DOI: 10.1109/JIOT.2022.3164478.

[25] J. Su, L. Zhang, and Y. Mu, 'BA-RMKABSE: Blockchain-aided ranked multi-keyword attribute-based searchable encryption with hiding policy for smart health system,' *Future Gener. Comput. Syst.*, vol. 132, pp. 299–309, 2022. DOI: 10.1016/j.future.2022.02.019.